The Directions of Technical Change

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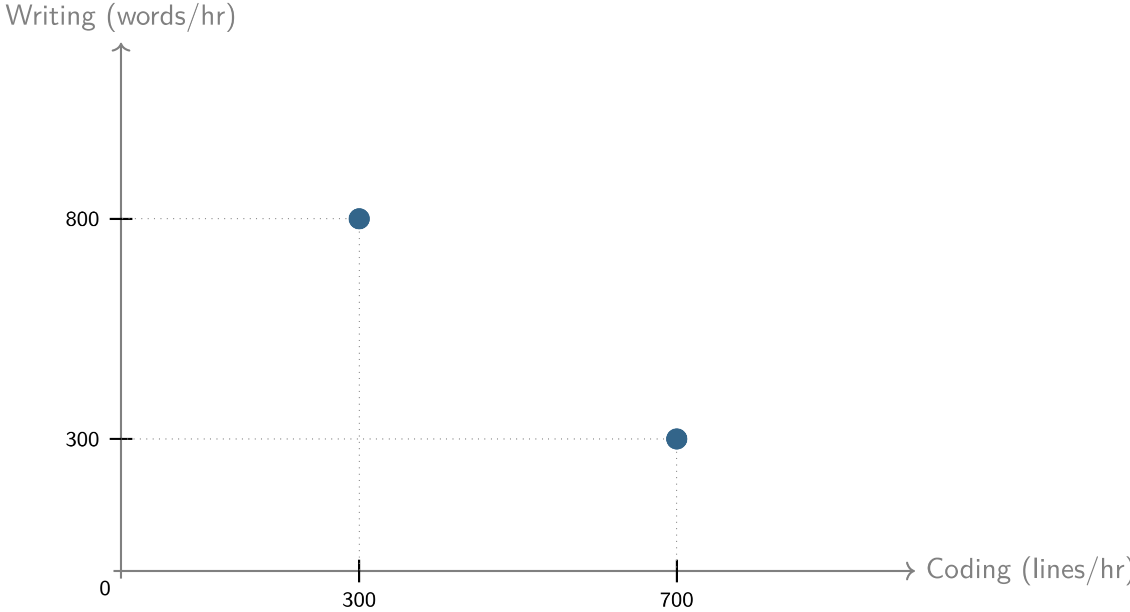
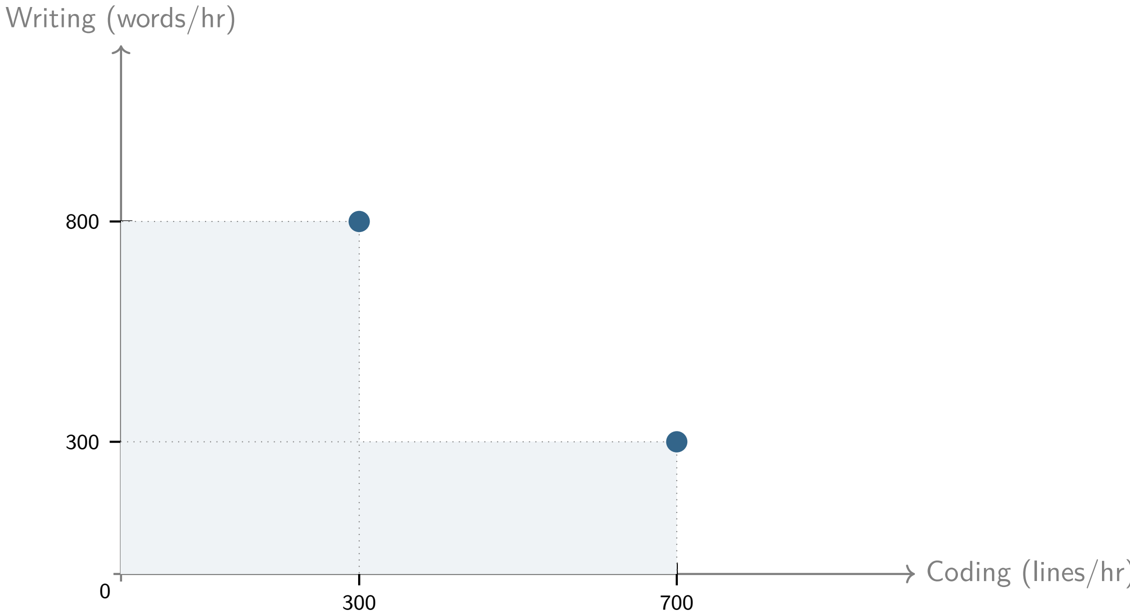
2026

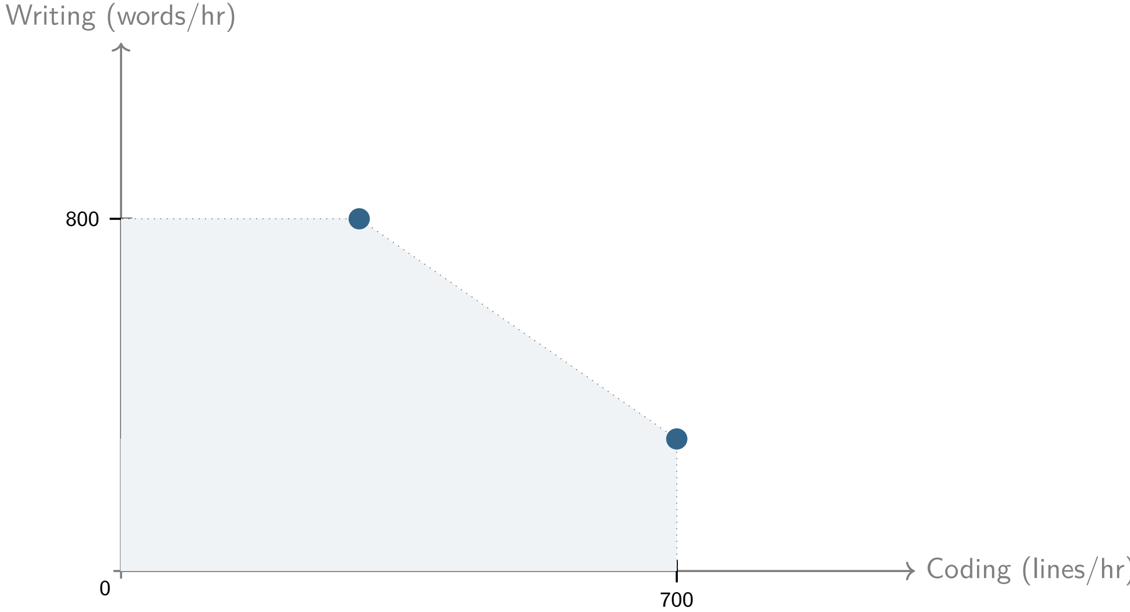
Suppose we observe a worker doing two tasks



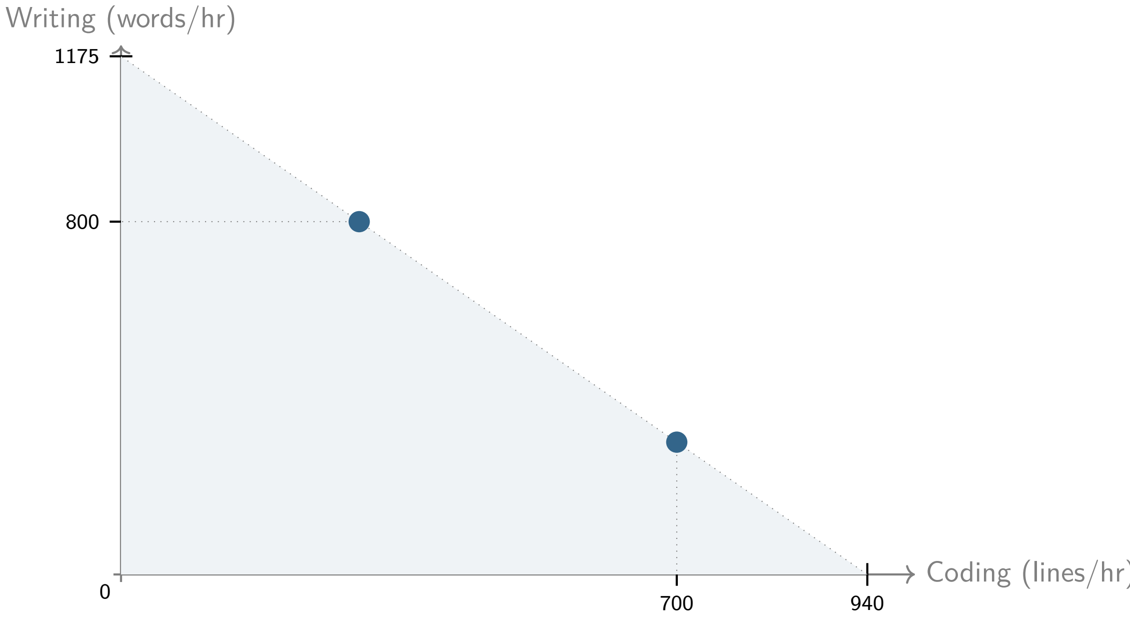
The worker can also do less: free disposal



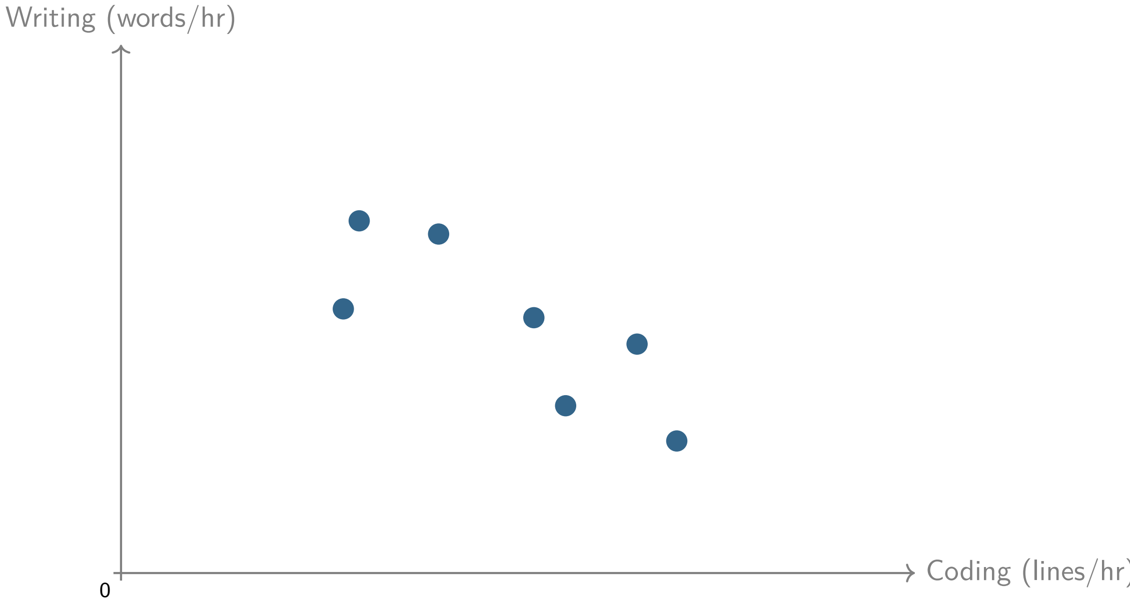
Time can be split across tasks: convex hull



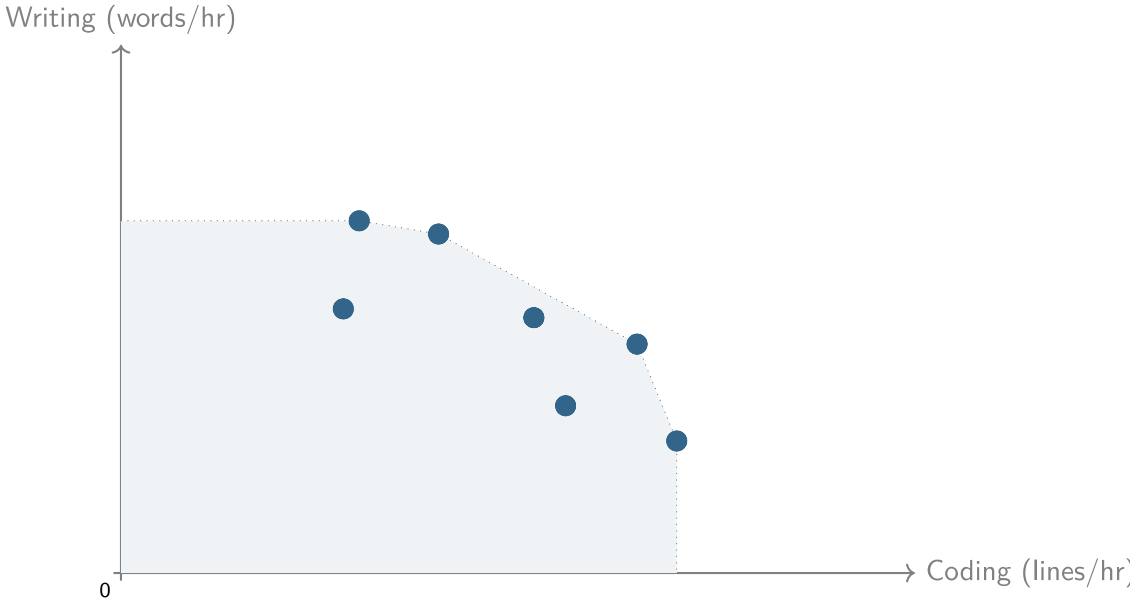
With perfect substitution: linear production possibility set (PPS)



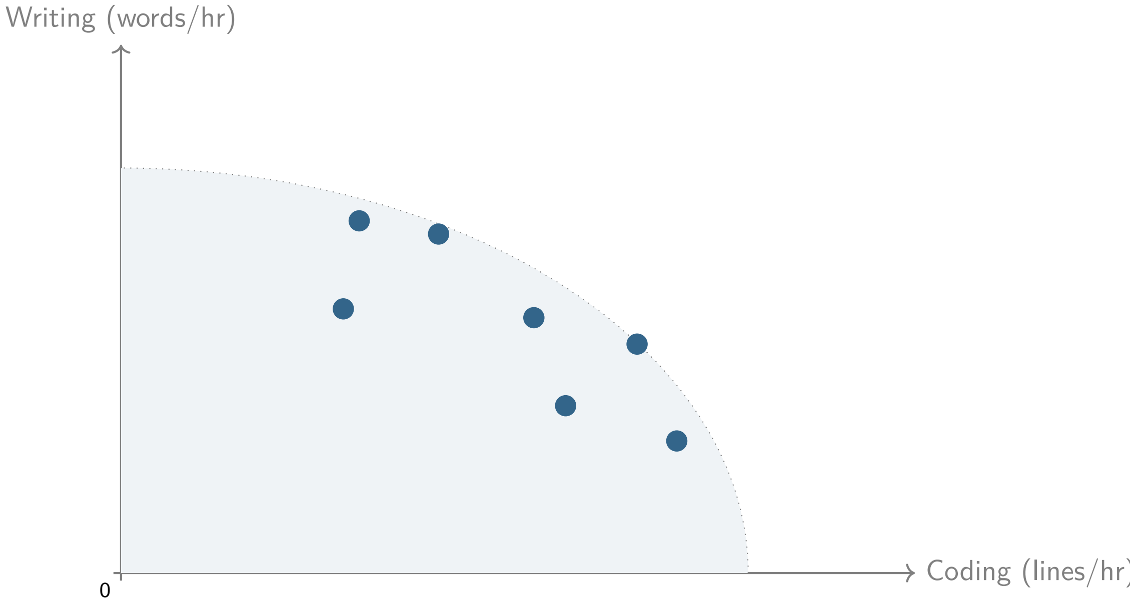
Many observed tasks for the same worker



PPS is the convex hull



Smooth approximation: constant-elasticity PPS



A model of AI adoption

Ingredients

1. Human capabilities bundled in a convex production possibility set (PPS)
2. AI capabilities limited by human supervision/attention time
3. (Flexible task allocation responsive to skills and job requirements)

The worker's problem

$$\max_x \; F(x) = \left(\sum_{i=1}^{N} \theta_i^{1/\sigma} x_i^{1-1/\sigma} \right)^{\sigma/(\sigma-1)}$$

subject to

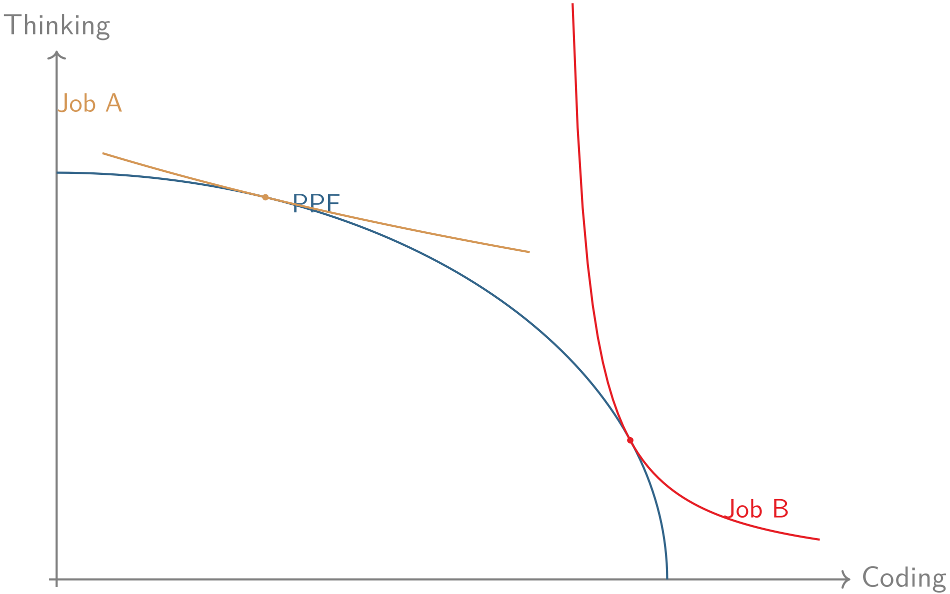
$$\left(\sum_{i=1}^{N} (x_i/s_i)^{1+1/\gamma} \right)^{\gamma/(\gamma+1)} \leq B$$

x = activities, θ = job requirements (CES), s = skills (CET), B = time in a day.

γ = elasticity of transformation (PPF curvature)

σ = elasticity of substitution (isoquant curvature)

Different jobs have different requirements



Optimal task mix depends on skills and requirements

$$x_i \propto s_i^{(1+\gamma)\sigma/(\gamma+\sigma)} \theta_i^{\gamma/(\gamma+\sigma)}$$

Productivity is a CES-weighted average of skills

Output per unit of time

$$\Phi = \left[\sum_k (s_k^{\sigma-1} \theta_k)^{(\gamma+1)/(\gamma+\sigma)} \right]^{(\gamma+\sigma)/[(\gamma+1)(\sigma-1)]}$$

Jevons' Paradox

Time share on activity i:

$$\omega_i^A = \frac{(s_i^{\sigma-1}\theta_i)^{(\gamma+1)/(\gamma+\sigma)}}{\sum_k (s_k^{\sigma-1}\theta_k)^{(\gamma+1)/(\gamma+\sigma)}}$$

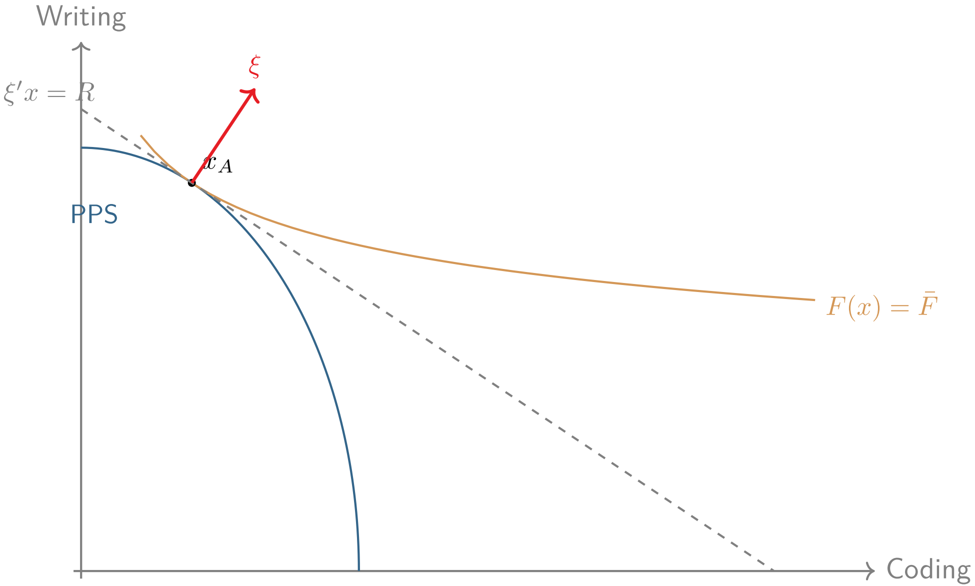
Increases in skill s_i if $\sigma > 1$. Relevant range for knowledge work.

Shadow prices express scarcity (marginal cost of human time)

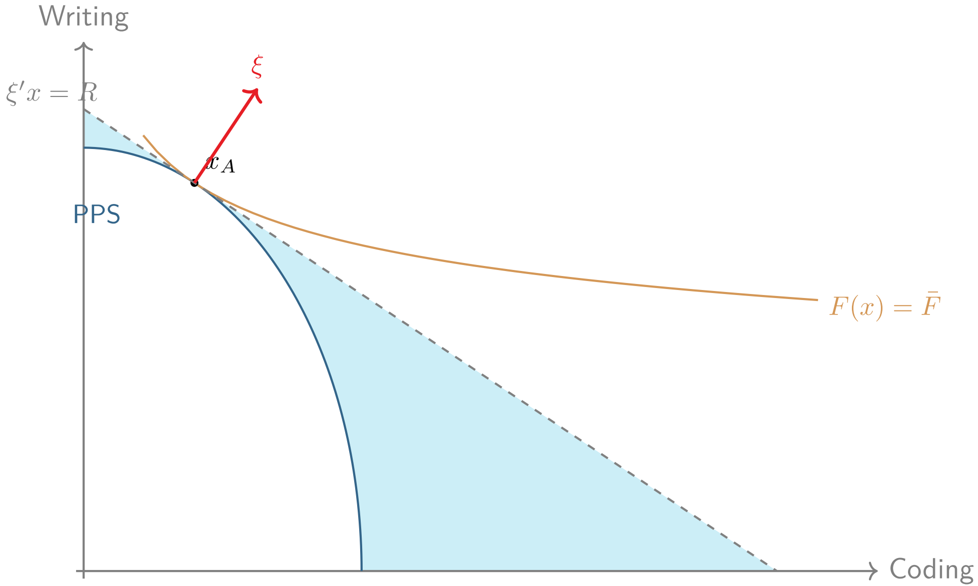
$$\xi_i = \left(\frac{\theta_i}{s_i^{1+\gamma}} \right)^{1/(\gamma+\sigma)},$$

High for activities that are required (θ_i high) but hard to do (s_i low).

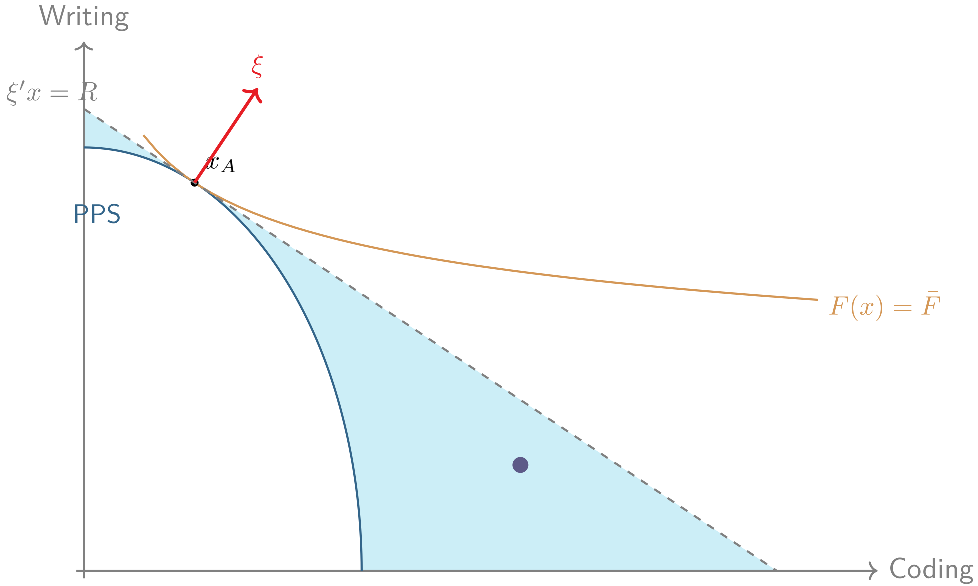
Shadow prices



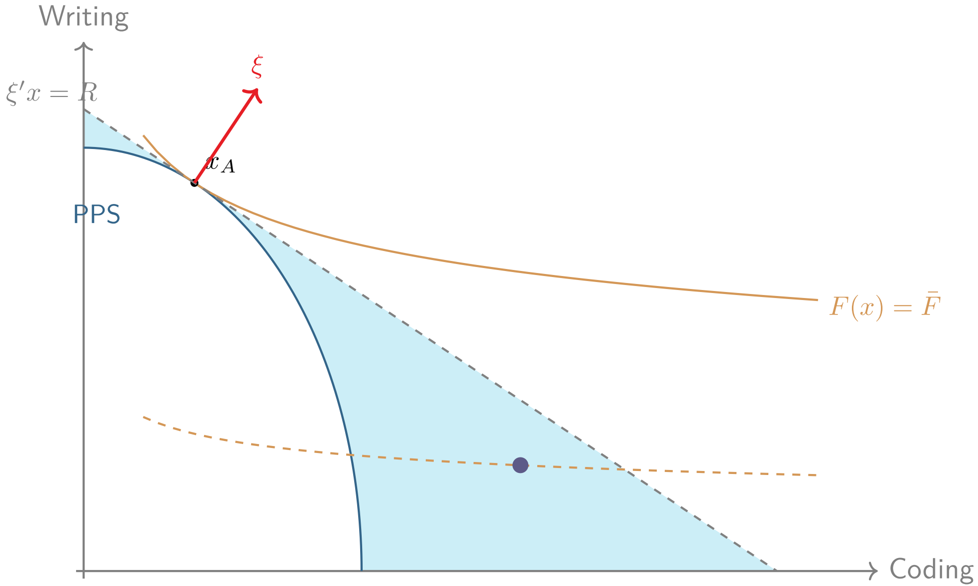
The budget set is larger than the PPS



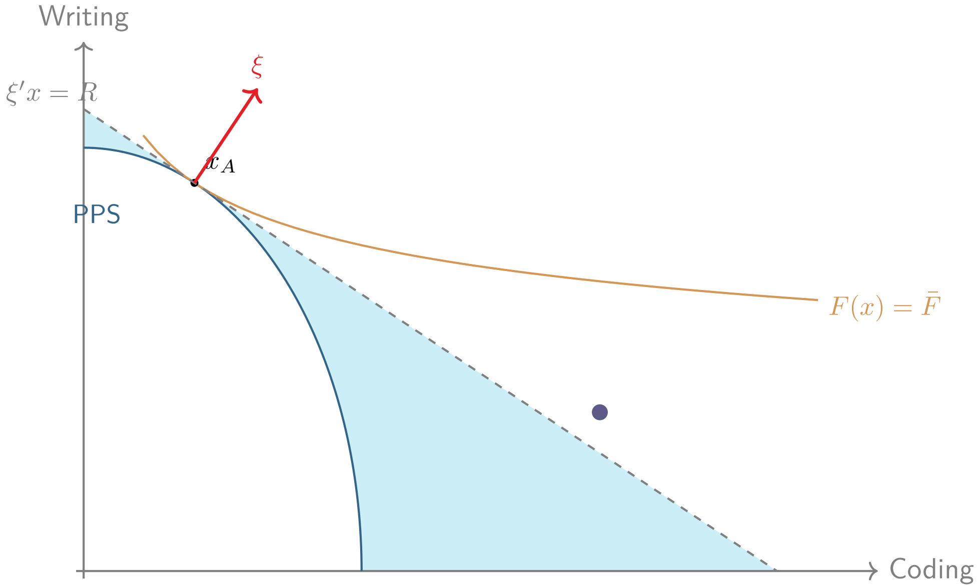
Suppose a machine can provide this amount of code and writing per hour



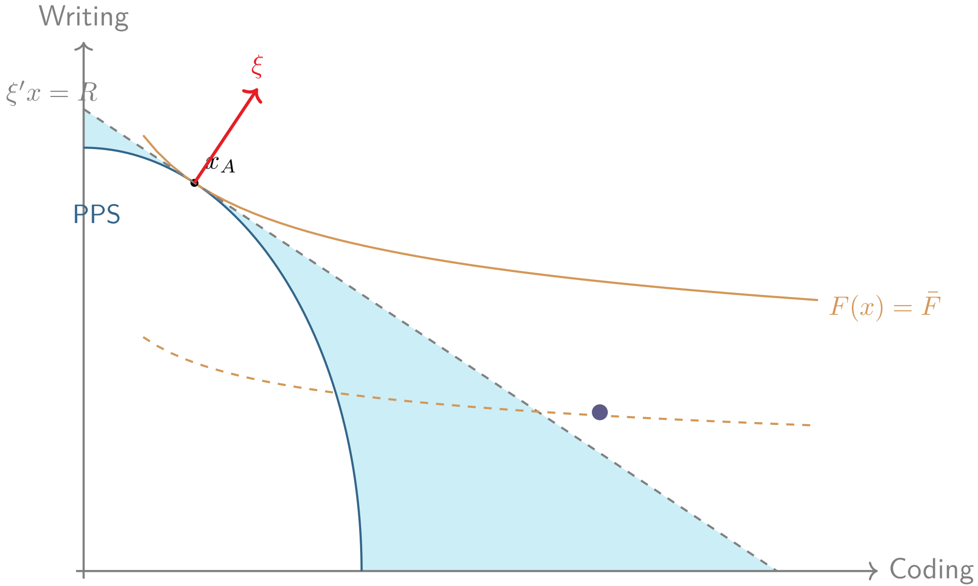
Would you buy it?



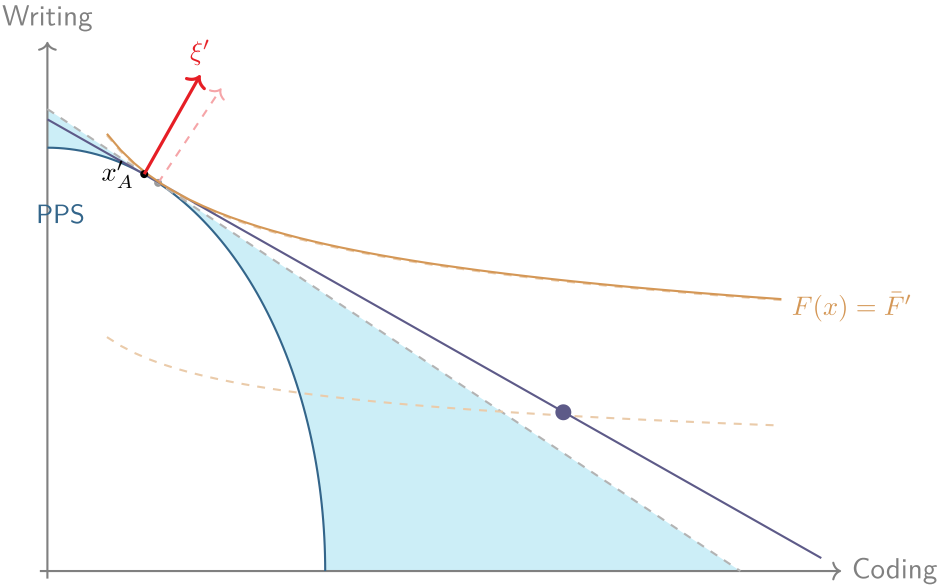
How about now?



AI alone is still worse than human alone



But AI + human now better



AI

What can AI do?

AI can perform any combination of $\{t_1, \ldots, t_N\}$ tasks, but requires supervision.

$$\max_{x, x_{\text{human}}, \mu} \; F(x)$$

$$x_i = x_{i,\text{human}} + \mu_i t_i$$

$$\sum_i \mu_i + g(x_{\text{human}}) \leq B$$

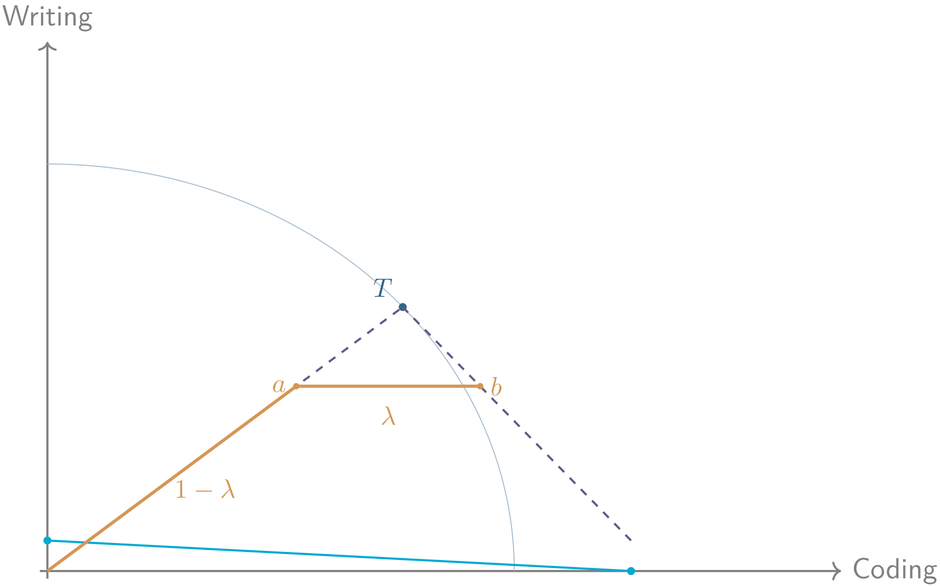
AI + human = convex hull

$$X_{AI} = \{x : \sum_i x_i/t_i \le \mu B\}$$

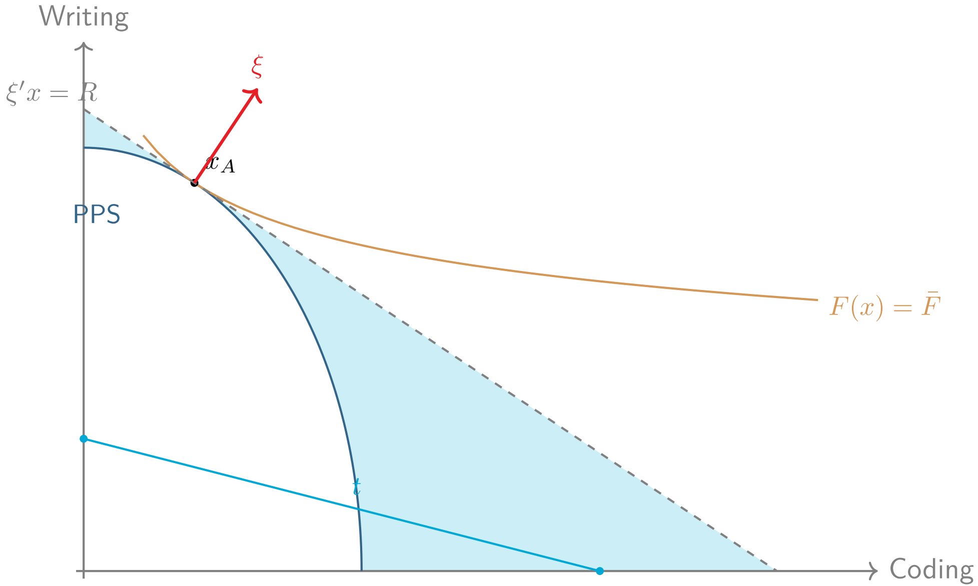
$$X_{\text{human}} = \{x : g(x) \le (1-\mu)B\}$$

$$X = X_{\text{human}} + X_{AI} = \{x : x = x_1 + x_2, x_1 \in X_{\text{human}}, x_2 \in X_{AI}\}$$

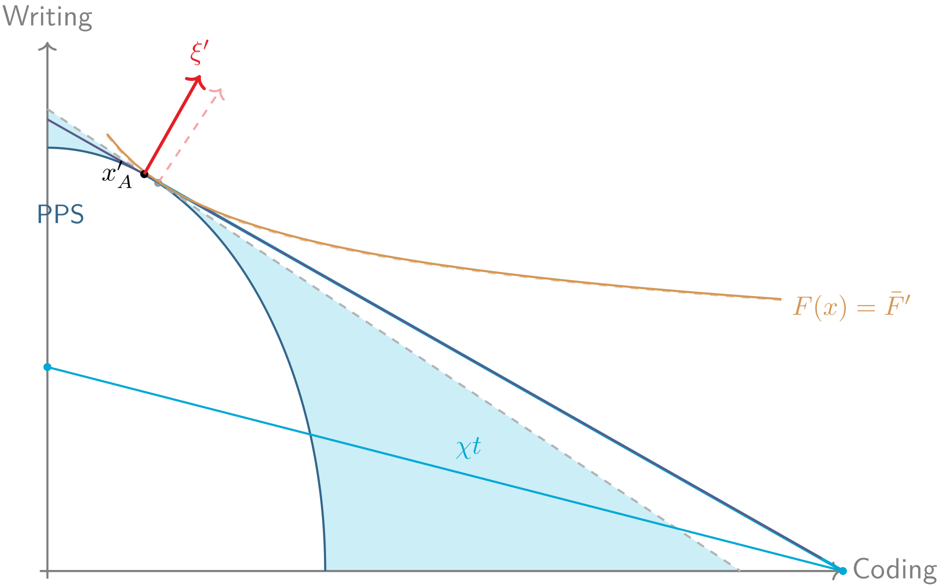
AI + human = convex hull



AI has an absolute but not comparative advantage



Scaling up: the budget line rotates



Adoption

When to adopt AI?

Whenever the value of AI tasks,

evaluated at the shadow prices of the human PPS,

exceeds your current output.

Adopt AI for first task i

$$t_i > \left(\frac{\gamma}{\gamma+1}\right)^{\gamma/(\gamma+1)} s_i \bar{\omega}_{oi}^{-1/(\gamma+1)}$$

$$\frac{t_i}{s_i} > \left(\frac{\gamma}{\gamma+1}\right)^{\gamma/(\gamma+1)} \bar{\omega}_{oi}^{-1/(\gamma+1)} > 1 \qquad \text{if } \gamma < \infty$$

Multitask adoption

Order tasks by highest $t_i/s_i \times \bar{\omega}_{oi}^{1/(\gamma+1)}$.

There exist $\chi_1 < \chi_2 < \cdots < \chi_N$ such that AI is used for the first k tasks if and only if

$$\chi_{k+1} \geq t_k/s_k \times \bar{\omega}_{ok}^{1/(\gamma+1)} \geq \chi_k$$

The general structure

All results depend on **convexity**, not on CES–CET:

- Any convex PPS $X_A = \{x : g(x) \leq B\}$ with quasi-convex hom(1) production function F
- Shadow prices ξ defined by tangency (supporting hyperplane)

CES–CET gives closed forms; the geometry works for any F and g.

Convexity is a reduced-form way to capture task complementarity within a fixed time budget.

Model predictions

1. Region of non-adoption: absolute advantage is not enough.
2. Fast adoption but small productivity effects past this threshold.
3. Intensive margin: Partial adoption even for augmented tasks
4. Different jobs augment+automate different tasks.

Shout-outs

Manuelli & Seshadri (2014)

Neoclassical theory explains slow adoption of tractors ($\varepsilon_{\text{horse,tractor}} < \infty$).

Garicano, Li and Wu (2026)

Human capabilities in *strong bundles* serve as bottlenecks to AI.

Freund and Mann (2026)

Automation reshapes the task mix of jobs ($\sigma = 1, \gamma = \infty$).

Jones (2025)

Complementarity (low σ) matters for progress more than benchmarks (t_i).

Empirical example

Data sources

Antropic Economic Index (AEI)

Task success rates, human times and AI times for O*NET tasks, March 2026. Aggregate to 332 Intermediate Work Activities (IWAs).

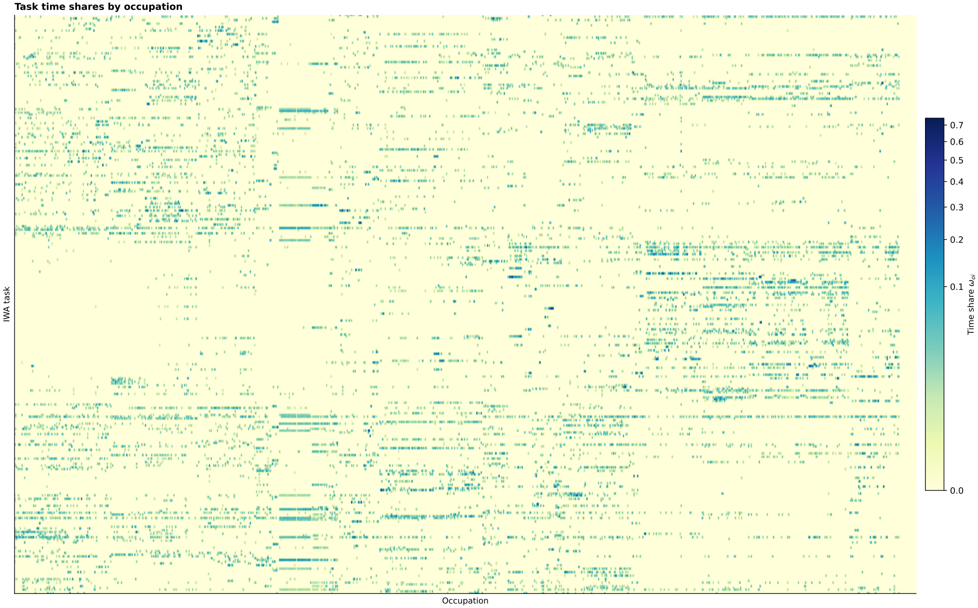
$$\tau_i = \frac{t_i}{s_i} = \frac{\text{human time}}{\text{AI time/success rate}}$$

O*NET occupational task frequencies

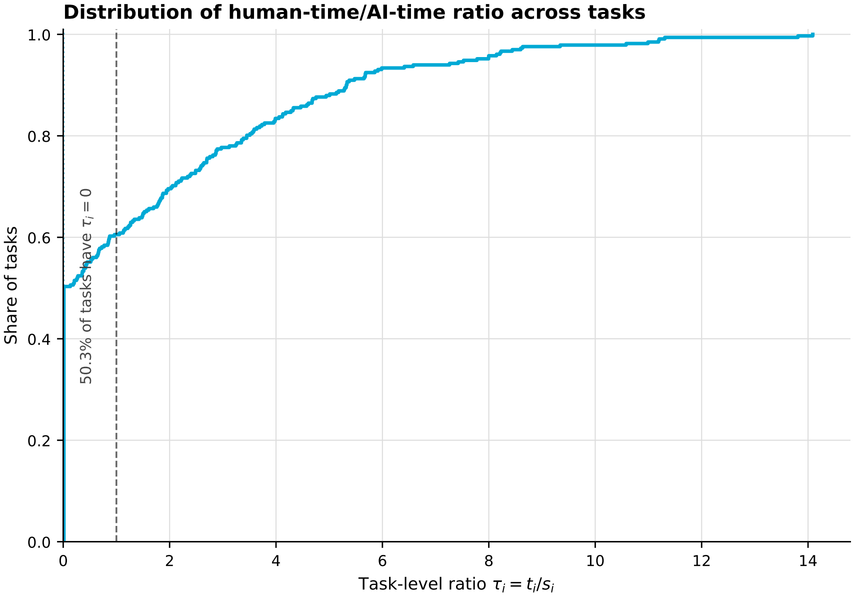
Use task frequnencies λ_{oi} (daily, weekly, etc) and approximate

$$\omega_{oi} \approx \frac{\lambda_{oi}}{\sum_j \lambda_{oj}}$$

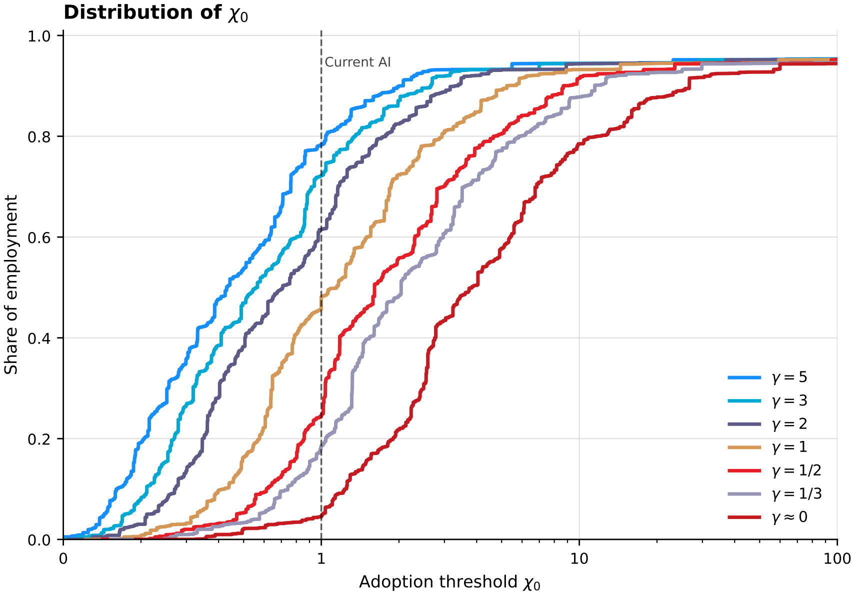
Task time shares across 1,016 occupations



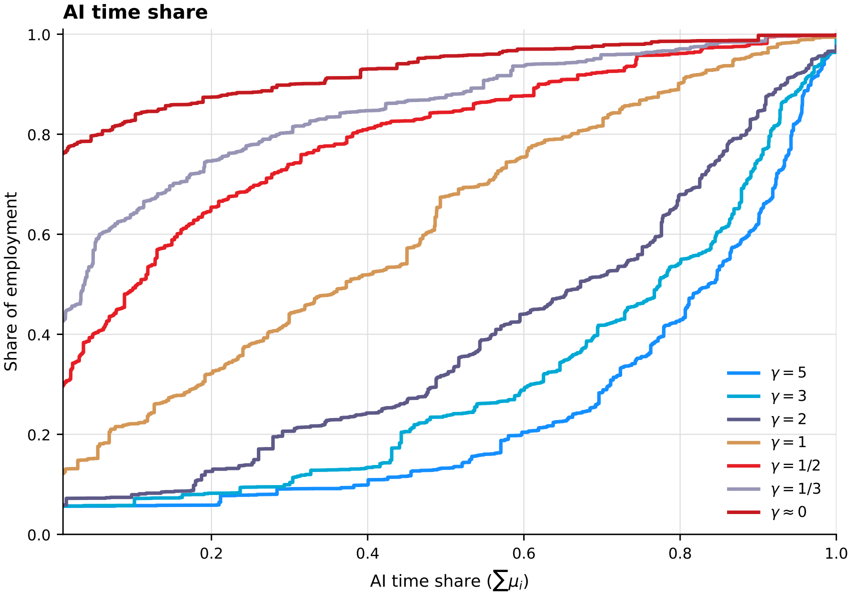
Task-level human-time/AI-time ratio



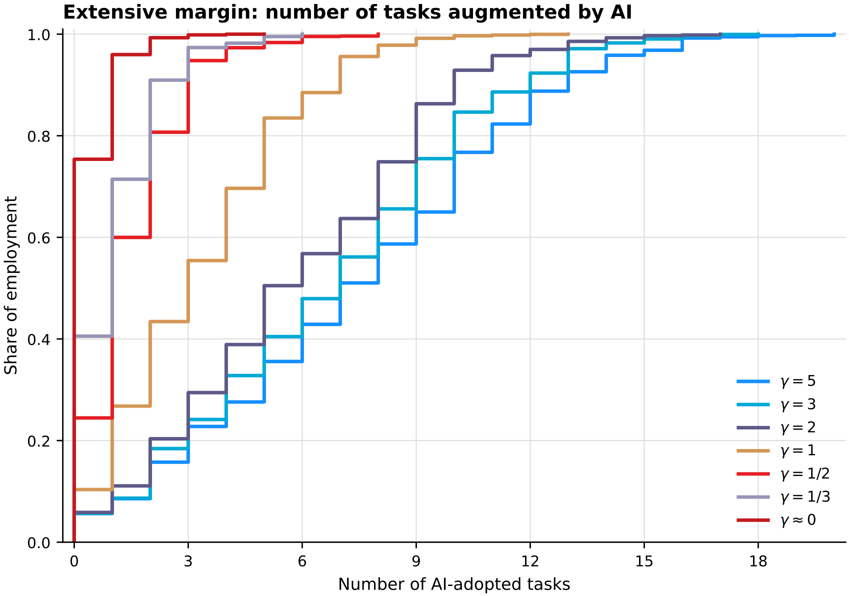
Distribution of adoption thresholds



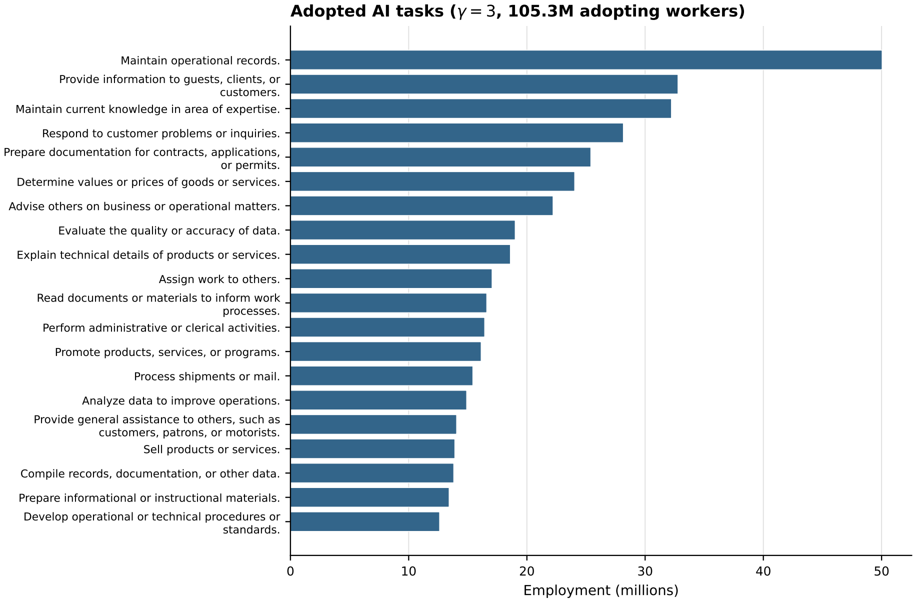
AI time share



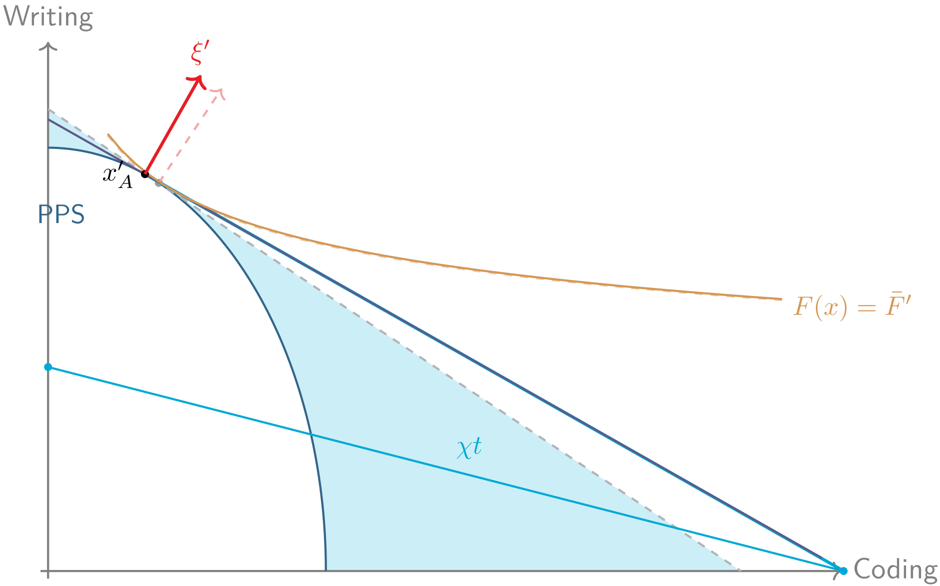
How many tasks does AI augment?



Which tasks get adopted first?



The figure to take away



Appendix

Acknowledgements

Funded by the European Union

European Research Council
Established by the European Commission

Nemzeti Kutatási, Fejlesztési és Innovációs Hivatal

AZ NKFI ALAPBÓL MEGVALÓSULÓ PROJEKT

This research was funded by the European Union under the Horizon Europe grant 101061123 and by the National Research, Development and Innovation Office (Forefront Research Excellence Program contract number 144193). Views and opinions expressed are those of the authors only and do not necessarily reflect those of the European Union, the European Commission, or the National Research, Development and Innovation Office.